Shiv Nadar University, Chennai
Department of Information Technology
25IT1731 Computational Thinking and Problem Solving with Python

Ex 7 – File Handling

Part A

1	Write a Python program to read and display the contents of a text file named ebook.txt
2	Define a function assignment_analyser() that accepts a filename and returns the total number of characters, words, and lines in assignment.txt.
3	Write a Python program to read the Python code stored in code.txt and copy it to recordcode.txt, excluding comment lines that begin with #.
4	A text file t1.txt contains alphanumeric characters. Open the file in a suitable mode that allows both reading and writing without closing it between operations. Append lines to the file, read the complete contents, and count the uppercase letters, lowercase letters, and digits.
5	Write a Python program to read code.txt and display the Python keywords present in the file using the keyword module.
6	Define a function display_words() that reads ebook.txt and returns a list of words containing fewer than four characters.
7	Define a function sort_studentname() that reads student names from student.txt, sorts them alphabetically, and writes the sorted names to sortstudent.txt.
8	Define a function Find_all() that accepts a filename and returns a dictionary containing the frequency of each word in the file.
9	A text file named secret.txt contains a secret message. Write a Python program to read the contents of secret.txt, reverse the order of the characters in the message, and write the reversed message to a new file named decoded.txt. Finally, read and display the contents of decoded.txt.
10	A text file named reviews.txt contains customer reviews, one review per line. For example: The product is excellent and easy to use. The quality is poor and disappointing. The delivery was fast and the service was good. The product is bad and expensive. The product is average. Use the following positive keywords: excellent, good, great, fast, easy, happy. Use the following negative keywords: bad, poor, slow, expensive, disappointing, worst. Write a Python program to read the file line by line and classify each review as Positive, Negative, or Neutral. Count positive and negative keywords without considering case. Classify a review as Positive when the positive count is greater, Negative when the negative count is greater, and Neutral when the counts are equal. Display each review with its classification, count the number of reviews in each category, and write the results to sentiment_report.txt.

Part B

11	A text file named attendance.txt stores student names and attendance percentages in the format Name,Percentage. The file contains the following records: Arun,92 Priya,68 Kumar,75 Divya,55 Ravi,85 Swathi,78 Classify attendance as Good (85% or above), Warning (75% or above but below 85%), or Low Attendance (below 75%). Write a Python program to read the records, classify each student, display the student name, attendance percentage, and category, and count the number of students in each category. Write the records of students with low attendance to attendance_alerts.txt.
12	A text file named environment.txt contains environmental reports, one report per line. For example: Heavy rainfall caused flooding in the city. Air pollution levels increased near the industrial area. The lake water quality improved after cleanup. Plastic waste was found along the beach. Tree planting reduced heat near the school. Use the following keyword categories: Water Issues: rainfall, flooding, lake, water Air Issues: air, pollution, smoke Waste Issues: plastic, waste, garbage Positive Environmental Action: improved, cleanup, planting, reduced Write a Python program to read the reports line by line and classify each report by matching keywords without considering case. Display each report and its category, and count the number of reports in each category. A report may belong to more than one category if it contains keywords from multiple categories. Write all reports classified under Water Issues, Air Issues, or Waste Issues to environment_alerts.txt.

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